Exploratory Data Analysis of Student Academic Performance Using Python

Data cleaning • Statistical analysis • Correlation • Visualization • Academic insights

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INTRODUCTION & OBJECTIVE

Exploratory Data Analysis (EDA) examines student records to understand data quality, distributions, relationships and performance patterns before predictive modeling.

OBJECTIVES

- Measure academic performance using Exam Score.
- Study the effect of attendance and study hours.
- Use correlation and visualization to discover patterns.

DATASET

Kaggle: Student Performance Factors

- 6,607 student records.
- 20 academic, behavioral and demographic variables.
- Main target: Exam Score.
- Key variables: Hours Studied, Attendance, Previous Scores, Sleep Hours and Tutoring Sessions.
- Categorical factors include Motivation Level and Parental Involvement.

Dataset source: Kaggle / Student Performance Factors

EDA WORKFLOW

1	Load	CSV dataset
2	Clean	Missing values & duplicates
3	Analyze	Mean, median, correlation
4	Visualize	Plots & heatmap
5	Interpret	Findings & recommendations

KEY CORRELATIONS

Attendance ↔ Exam Score	0.58
Hours Studied ↔ Exam Score	0.45
Previous Scores ↔ Exam Score	0.18
Tutoring ↔ Exam Score	0.16
Sleep ↔ Exam Score	−0.02

RESULTS & VISUALIZATION

Figure: Exam-score distribution, study-hours relationship, attendance relationship and correlation matrix

MAJOR FINDINGS

- Attendance has the strongest shown correlation with Exam Score (r = 0.58).
- Study hours also show a positive relationship (r = 0.45).
- Previous scores and tutoring have weaker positive relationships.

INTERPRETATION

- The scatter plots show positive trends rather than perfect prediction.
- The score histogram shows most observations concentrated around the mid-to-high 60s.

CONCLUSION & FUTURE SCOPE

- EDA reveals important academic patterns and supports evidence-based decisions.
- Next: predict performance using Regression, Decision Tree and Random Forest.